

ADER–DG Formulation for the Exner Model in the Low-Froude Regime

1 The shallow-water–Exner model in the low-Froude regime

We consider the one-dimensional shallow-water–Exner model, with the shallow-water part written in the low-Froude-number regime. Let $\text{Fr} > 0$ denote the Froude number, defined from characteristic velocity and water-depth scales by

$$\text{Fr} = \frac{U_{\text{ref}}}{\sqrt{gH_{\text{ref}}}}.$$

In the following, we are interested in the regime $\text{Fr} \ll 1$. With the standard nondimensional low-Froude scaling, the gravitational terms in the shallow-water momentum equation are multiplied by $1/\text{Fr}^2$. The Exner equation and the sediment-transport law are kept unchanged.

Let $h = h(x, t)$ denote the water height, $u = u(x, t)$ the depth-averaged water velocity, $q = q(x, t) = h(x, t)u(x, t)$ the water discharge, and $b = b(x, t)$ the evolving bottom topography. The free-surface elevation is

$$\eta = h + b,$$

so that

$$h = \eta - b, \quad u = \frac{q}{h} = \frac{q}{\eta - b}$$

whenever $h > 0$.

The sediment-transport discharge is described by the classical Grass law

$$q_b = \xi A u |u|^{m_g - 1}, \quad \xi = \frac{1}{1 - \rho_0}, \quad 0 \leq A \leq 1, \quad m_g = 3,$$

where $\rho_0 \in [0, 1)$ is the porosity of the sediment layer. The empirical coefficient A accounts for effects such as grain size and kinematic viscosity; larger values of A correspond to stronger fluid–sediment interaction.

The water mass equation and the Exner equation are, respectively,

$$\partial_t h + \partial_x q = 0, \quad \partial_t b + \partial_x q_b = 0.$$

Since $\eta = h + b$, adding these two equations gives

$$\partial_t \eta + \partial_x (q + q_b) = 0.$$

Moreover, the momentum balance can be written in the pre-balanced form

$$\partial_t q + \partial_x \left(qu + \frac{1}{2\text{Fr}^2} (\eta^2 - 2\eta b) \right) = -\frac{\eta}{\text{Fr}^2} \partial_x b.$$

Consequently, using the variables (η, q, b) , the shallow-water–Exner system takes the compact form

$$\partial_t \mathbf{U}(x, t) + \partial_x \mathbf{F}(\mathbf{U}(x, t)) = \mathbf{S}(x, t), \tag{6}$$

where

$$\mathbf{U} = \begin{pmatrix} \eta \\ q \\ b \end{pmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{pmatrix} q + q_b \\ qu + \frac{1}{2\text{Fr}^2}(\eta^2 - 2\eta b) \\ q_b \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 0 \\ -\frac{\eta}{\text{Fr}^2} \partial_x b \\ 0 \end{pmatrix}.$$

The third component q_b of the flux is required by the Exner equation $\partial_t b + \partial_x q_b = 0$.

2 ADER–DG formulation

Unlike a Runge–Kutta DG discretization, the ADER–DG method uses the ADER approach for the temporal evolution. The resulting method is one-step, one-stage, and fully discrete.

2.1 Mesh and approximation space

Let the spatial domain be partitioned into N cells

$$I_j = [x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}], \quad j = 1, \dots, N,$$

with cell centers and cell sizes

$$x_j = \frac{1}{2} (x_{j-\frac{1}{2}} + x_{j+\frac{1}{2}}), \quad \tau_j = x_{j+\frac{1}{2}} - x_{j-\frac{1}{2}}, \quad \tau = \max_{1 \leq j \leq N} \tau_j.$$

For the time interval $[t^n, t^{n+1}]$, define the local space–time control volume

$$\Omega_j^n = I_j \times [t^n, t^{n+1}].$$

The discontinuous finite-element space of degree at most k is

$$V_h^k = \left\{ v : v|_{I_j} \in \mathbb{P}^k(I_j), \quad j = 1, \dots, N \right\}.$$

Thus, the DG approximation satisfies $\mathbf{U}_h(\cdot, t) \in [V_h^k]^3$. Let $\phi_\ell = \phi_\ell(x) \in V_h^k$, $\ell = 0, \dots, k$, be an arbitrary time-independent test function.

2.2 Space–time weak formulation

Multiplying system (6) by ϕ_ℓ , integrating over Ω_j^n , and integrating the temporal and spatial derivatives by parts gives

$$\begin{aligned} & \int_{I_j} \mathbf{U}_{h,j}^{n+1}(x) \phi_\ell(x) dx - \int_{I_j} \mathbf{U}_{h,j}^n(x) \phi_\ell(x) dx - \int_{t^n}^{t^{n+1}} \int_{I_j} \mathbf{F}(\mathbf{U}_{\tau,j}(x, t)) \partial_x \phi_\ell(x) dx dt \\ & + \widehat{\mathbf{F}}_{j+\frac{1}{2}}^n \phi_\ell(x_{j+\frac{1}{2}}^-) - \widehat{\mathbf{F}}_{j-\frac{1}{2}}^n \phi_\ell(x_{j-\frac{1}{2}}^+) = \int_{t^n}^{t^{n+1}} \int_{I_j} \mathbf{S}_{\tau,j}(x, t) \phi_\ell(x) dx dt. \end{aligned} \quad (7)$$

This vector identity holds for $j = 1, \dots, N$. Here, $\widehat{\mathbf{F}}_{j+\frac{1}{2}}^n$ and $\widehat{\mathbf{F}}_{j-\frac{1}{2}}^n$ denote the time-integrated numerical flux contributions at the right and left interfaces of I_j , respectively; they may be constructed from a consistent Lax–Friedrichs numerical flux. Their explicit computation is not considered at this stage.

2.3 Initialization and local space–time prediction

Step 1: Initialization. Let x_j denote the center of the cell I_j . The initial DG approximation in I_j is written as

$$\mathbf{U}_{h,j}^0(x) = \mathbf{U}_{\tau,j}(x, 0) = \sum_{k_x=0}^k \tilde{\mathbf{U}}_j(k_x, 0)(x - x_j)^{k_x}, \quad x \in I_j. \quad (9)$$

The coefficients $\tilde{\mathbf{U}}(k_x, 0)$, $k_x = 0, \dots, k$, are obtained componentwise by the L^2 -projection of the initial condition $\mathbf{U}_0(x) = \mathbf{U}(x, 0)$ onto $\mathbb{P}^k(I_j)$. More precisely, they satisfy

$$\int_{I_j} (\mathbf{U}_{\tau}(x, 0) - \mathbf{U}_0(x)) \phi_{\ell}(x) dx = \mathbf{0}, \quad \ell = 0, \dots, k.$$

Step 2: Differential transformation procedure. Instead of employing the Cauchy–Kowalevski procedure used in traditional ADER methods, we adopt the differential transformation (DT) procedure. Suppose that the DG polynomial $\mathbf{U}_{h,j}^n(x)$ is known in the cell I_j . The local space–time approximation is expressed as the truncated Taylor expansion

$$\mathbf{U}_{\tau}(x, t) = \sum_{k_t=0}^k \sum_{k_x=0}^{k-k_t} \tilde{\mathbf{U}}(k_x, k_t)(x - x_j)^{k_x} (t - t^n)^{k_t}, \quad (x, t) \in \Omega_j^n, \quad (10a)$$

where

$$\tilde{\mathbf{U}}(k_x, k_t) = \frac{1}{k_x! k_t!} \left. \frac{\partial^{k_x+k_t} \mathbf{U}(x, t)}{\partial x^{k_x} \partial t^{k_t}} \right|_{(x,t)=(x_j,t^n)} \quad (10b)$$

denotes the differential transform of $\mathbf{U}$ at (x_j, t^n) . The cell and time-step indices are omitted from the transformed coefficients for simplicity.

The differential transformation rules needed in the present study are given below. For arbitrary functions $v(x, t)$, $z(x, t)$, and a constant c , one has

$$\begin{aligned} w(x, t) = c v(x, t) &\implies \tilde{w}(k_x, k_t) = c \tilde{v}(k_x, k_t), \\ w(x, t) = \partial_x v(x, t) &\implies \tilde{w}(k_x, k_t) = (k_x + 1) \tilde{v}(k_x + 1, k_t), \\ w(x, t) = \partial_t v(x, t) &\implies \tilde{w}(k_x, k_t) = (k_t + 1) \tilde{v}(k_x, k_t + 1), \\ w(x, t) = v(x, t) z(x, t) &\implies \tilde{w}(k_x, k_t) = \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{v}(r, s) \tilde{z}(k_x - r, k_t - s). \end{aligned}$$

For $w(x, t) = 1/v(x, t)$, assuming $\tilde{v}(0, 0) \neq 0$, the transformed coefficients are determined recursively by

$$\begin{cases} \tilde{w}(0, 0) = \frac{1}{\tilde{v}(0, 0)}, \\ \tilde{w}(k_x, k_t) = -\frac{1}{\tilde{v}(0, 0)} \sum_{\substack{0 \leq r \leq k_x, 0 \leq s \leq k_t \\ r+s > 0}} \tilde{v}(r, s) \tilde{w}(k_x - r, k_t - s), \quad (k_x, k_t) \neq (0, 0). \end{cases}$$

Applying the DT procedure componentwise to

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}(\mathbf{U}) = \mathbf{S}$$

gives

$$\tilde{\mathbf{U}}(k_x, k_t + 1) = -\frac{k_x + 1}{k_t + 1} \tilde{\mathbf{F}}(k_x + 1, k_t) + \frac{1}{k_t + 1} \tilde{\mathbf{S}}(k_x, k_t). \quad (11)$$

This recurrence is applied for

$$k_t = 0, \dots, k-1, \quad k_x = 0, \dots, k-k_t-1.$$

For the low-Froude shallow-water-Exner system considered here, the transformed flux and source are decomposed as

$$\left\{ \begin{array}{l} \tilde{\mathbf{F}}(k_x+1, k_t) = \begin{pmatrix} \tilde{Q}_1(k_x+1, k_t) \\ \tilde{G}_1(k_x+1, k_t) + \tilde{G}_2(k_x+1, k_t) - \tilde{G}_3(k_x+1, k_t) \\ \tilde{Q}_2(k_x+1, k_t) \end{pmatrix}, \\ \tilde{\mathbf{S}}(k_x, k_t) = \begin{pmatrix} 0 \\ -\tilde{G}_4(k_x, k_t) \\ 0 \end{pmatrix}. \end{array} \right. \quad (12)$$

We next express these intermediate quantities in terms of the transformed coefficients of the main variables (η, q, b) . Since

$$h = \eta - b, \quad u = \frac{q}{h},$$

we first define

$$\tilde{h}(k_x, k_t) = \tilde{\eta}(k_x, k_t) - \tilde{b}(k_x, k_t).$$

Let $R = 1/h$. Assuming $\tilde{h}(0, 0) > 0$, its transformed coefficients satisfy

$$\left\{ \begin{array}{l} \tilde{R}(0, 0) = \frac{1}{\tilde{h}(0, 0)}, \\ \tilde{R}(k_x, k_t) = -\frac{1}{\tilde{h}(0, 0)} \sum_{\substack{0 \leq r \leq k_x, 0 \leq s \leq k_t \\ r+s > 0}} \tilde{h}(r, s) \tilde{R}(k_x - r, k_t - s). \end{array} \right.$$

Consequently,

$$\tilde{u}(k_x, k_t) = \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{q}(r, s) \tilde{R}(k_x - r, k_t - s).$$

Since $m_g = 3$, the Grass sediment transport law becomes

$$q_b = \xi A u |u|^2 = \xi A u^3.$$

Introducing

$$\tilde{u}^2(k_x, k_t) = \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{u}(r, s) \tilde{u}(k_x - r, k_t - s),$$

the transformed sediment discharge is

$$\begin{aligned} \tilde{Q}_2(k_x, k_t) &= \tilde{q}_b(k_x, k_t) \\ &= \xi A \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{u}^2(r, s) \tilde{u}(k_x - r, k_t - s). \end{aligned}$$

Therefore, the first flux component is

$$\tilde{Q}_1(k_x, k_t) = \tilde{q}(k_x, k_t) + \tilde{Q}_2(k_x, k_t).$$

The transformed coefficients associated with the momentum flux are

$$\begin{aligned}
\tilde{G}_1(k_x, k_t) &= (\widetilde{qu})(k_x, k_t) \\
&= \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{q}(r, s) \tilde{u}(k_x - r, k_t - s), \\
\tilde{G}_2(k_x, k_t) &= \frac{1}{2\text{Fr}^2} (\widetilde{\eta^2})(k_x, k_t) \\
&= \frac{1}{2\text{Fr}^2} \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{\eta}(r, s) \tilde{\eta}(k_x - r, k_t - s), \\
\tilde{G}_3(k_x, k_t) &= \frac{1}{\text{Fr}^2} (\widetilde{\eta b})(k_x, k_t) \\
&= \frac{1}{\text{Fr}^2} \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{\eta}(r, s) \tilde{b}(k_x - r, k_t - s).
\end{aligned}$$

Finally, since the topography $b = b(x, t)$ evolves in time, we have

$$\tilde{b}_x(k_x, k_t) = (k_x + 1) \tilde{b}(k_x + 1, k_t).$$

It follows that

$$\begin{aligned}
\tilde{G}_4(k_x, k_t) &= \frac{1}{\text{Fr}^2} (\widetilde{\eta b_x})(k_x, k_t) \\
&= \frac{1}{\text{Fr}^2} \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{\eta}(r, s) \tilde{b}_x(k_x - r, k_t - s) \\
&= \frac{1}{\text{Fr}^2} \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} (k_x - r + 1) \tilde{\eta}(r, s) \tilde{b}(k_x - r + 1, k_t - s).
\end{aligned}$$

Starting from the spatial expansion coefficients $\tilde{\mathbf{U}}(k_x, 0)$, $k_x = 0, \dots, k$, the recurrence relation (11), together with the transformed expressions in (12), is applied successively in increasing order of k_t . This procedure determines all the space-time expansion coefficients

$$\tilde{\mathbf{U}}(k_x, k_t), \quad \tilde{\mathbf{F}}(k_x, k_t), \quad \tilde{\mathbf{S}}(k_x, k_t), \quad k_t = 0, \dots, k, \quad k_x = 0, \dots, k - k_t.$$

$$\begin{aligned}
\mathbf{U}_{\tau,j}(x, t) &= \sum_{k_t=0}^k \sum_{k_x=0}^{k-k_t} \tilde{\mathbf{U}}_j(k_x, k_t) (x - x_j)^{k_x} (t - t^n)^{k_t}, \\
\mathbf{F}_{\tau,j}(x, t) &= \sum_{k_t=0}^k \sum_{k_x=0}^{k-k_t} \tilde{\mathbf{F}}_j(k_x, k_t) (x - x_j)^{k_x} (t - t^n)^{k_t}, \\
\mathbf{S}_{\tau,j}(x, t) &= \sum_{k_t=0}^k \sum_{k_x=0}^{k-k_t} \tilde{\mathbf{S}}_j(k_x, k_t) (x - x_j)^{k_x} (t - t^n)^{k_t}.
\end{aligned}$$

for $(x, t) \in \Omega_j^n$, $j = 1, \dots, N$. In this manner, the local space-time evolution is obtained independently within each space-time control volume Ω_j^n .

Remark. The source coefficients $\tilde{\mathbf{S}}(k_x, k_t)$ are required for $k_t = 0, \dots, k-1$, $k_x = 0, \dots, k - k_t - 1$, which is precisely the range needed in recurrence (11).

2.4 Implementation steps of the ADER–DG method

For clarity, the implementation of the proposed ADER–DG method over one time step $t^n \longrightarrow t^{n+1}$ can be summarized as follows.

1. Initialization.

At $t = 0$, compute the initial DG polynomial in every cell I_j , $j = 1, \dots, N$, by the componentwise L^2 -projection of the initial data $\mathbf{U}_0(x) = \mathbf{U}(x, 0)$ onto $[\mathbb{P}^k(I_j)]^3$:

$$\mathbf{U}_{h,j}^0(x) = \sum_{k_x=0}^k \tilde{\mathbf{U}}_j(k_x, 0)(x - x_j)^{k_x}.$$

At a general time t^n , the spatial coefficients $\tilde{\mathbf{U}}_j(k_x, 0)$ are obtained from the known DG polynomial $\mathbf{U}_{h,j}^n(x)$.

2. Local space–time prediction.

Starting from the coefficients $\tilde{\mathbf{U}}_j(k_x, 0)$, apply the recurrence relation (11), together with the transformed flux and source expressions (12), successively in increasing order of k_t . This gives

$$\tilde{\mathbf{U}}_j(k_x, k_t), \quad \tilde{\mathbf{F}}_j(k_x, k_t), \quad \tilde{\mathbf{S}}_j(k_x, k_t),$$

for

$$k_t = 0, \dots, k, \quad k_x = 0, \dots, k - k_t.$$

The corresponding local space–time predictors $\mathbf{U}_{\tau,j}$, $\mathbf{F}_{\tau,j}$, and $\mathbf{S}_{\tau,j}$ are then reconstructed over $\Omega_j^n = I_j \times [t^n, t^{n+1}]$. This predictor step is performed independently in every cell.

3. Evaluation of the interface fluxes.

At each interface $x_{j+\frac{1}{2}}$, evaluate the left and right predictor traces

$$\mathbf{U}_{\tau,j}(x_{j+\frac{1}{2}}^-, t), \quad \mathbf{U}_{\tau,j+1}(x_{j+\frac{1}{2}}^+, t),$$

and use them to construct a consistent well-balanced numerical flux

$$\mathbf{F}_{j+\frac{1}{2}}^*(t) = \mathbf{F}^*\left(\mathbf{U}_{\tau,j}(x_{j+\frac{1}{2}}^-, t), \mathbf{U}_{\tau,j+1}(x_{j+\frac{1}{2}}^+, t)\right).$$

Its time-integrated contribution is

$$\hat{\mathbf{F}}_{j+\frac{1}{2}}^n = \int_{t^n}^{t^{n+1}} \mathbf{F}_{j+\frac{1}{2}}^*(t) dt.$$

The well-balanced interface flux must be coupled with a compatible discretization of the source term so that the targeted equilibrium state is preserved exactly.

4. Evaluation of the space–time integrals.

Evaluate all the volume, source, and interface integrals appearing in the weak formulation (7). In practice, these integrals are computed using sufficiently accurate Gaussian quadrature rules in both space and time. The quadrature order must be selected so that the polynomial terms involved in the ADER–DG discretization are integrated with the required accuracy.

5. One-stage DG update.

For each test function ϕ_ℓ , $\ell = 0, \dots, k$, compute the DG solution at t^{n+1} from the one-stage formula

$$\begin{aligned} \int_{I_j} \mathbf{U}_{h,j}^{n+1}(x) \phi_\ell(x) dx &= \int_{I_j} \mathbf{U}_{h,j}^n(x) \phi_\ell(x) dx + \int_{t^n}^{t^{n+1}} \int_{I_j} \mathbf{F}_{\tau,j}(x, t) \partial_x \phi_\ell(x) dx dt \\ &\quad - \widehat{\mathbf{F}}_{j+\frac{1}{2}}^n \phi_\ell(x_{j+\frac{1}{2}}^-) + \widehat{\mathbf{F}}_{j-\frac{1}{2}}^n \phi_\ell(x_{j-\frac{1}{2}}^+) \\ &\quad + \int_{t^n}^{t^{n+1}} \int_{I_j} \mathbf{S}_{\tau,j}(x, t) \phi_\ell(x) dx dt. \end{aligned}$$

The degrees of freedom of $\mathbf{U}_{h,j}^{n+1}$ are obtained by solving the resulting local mass-matrix system.

6. Advance to the next time step.

Set $n \leftarrow n + 1$, determine the next time-step size according to the appropriate CFL restriction, and repeat Steps 2–5 until the final time is reached.

Remark on nonlinear stabilization. A monotone interface flux, such as the local Lax–Friedrichs or Rusanov flux, provides numerical dissipation at cell interfaces, but it does not, by itself, prevent spurious oscillations in the higher-order DG polynomial near shocks or strong discontinuities. Therefore, for shock-dominated solutions, a TVB, WENO, or another suitable nonlinear limiter should be used. The limiter is applied to the updated spatial DG polynomial $\mathbf{U}_{h,j}^{n+1}$, preferably only in cells detected as troubled, and not directly to the local space–time predictor $\mathbf{U}_{\tau,j}(x, t)$. The limiting procedure must preserve the cell average and should be designed consistently with the well-balanced property. As an alternative for very strong shocks, a troubled-cell ADER–DG method with a robust first-order finite-volume numerical fluxes where needed may be employed.

2.5 DG Approximation and Local Space–Time Predictor

Throughout the following presentation, we distinguish between the spatial DG approximation and the local space–time predictor. At the discrete time t^n , the approximate DG solution is denoted by $\mathbf{U}_h^n(x)$, whose restriction

$$\mathbf{U}_{h,j}^n(x) = \mathbf{U}_h^n(x)|_{I_j}$$

is a polynomial of degree at most k in the spatial variable.

Starting from $\mathbf{U}_{h,j}^n$, the differential transformation procedure constructs, independently in each cell I_j , a local space–time predictor $\mathbf{U}_{\tau,j}(x, t)$ over $I_j \times [t^n, t^{n+1}]$. This predictor is initialized by the known DG polynomial:

$$\mathbf{U}_{\tau,j}(x, t^n) = \mathbf{U}_{h,j}^n(x), \quad x \in I_j. \quad (1)$$

Indeed, the local predictor is written as

$$\mathbf{U}_{\tau,j}(x, t) = \sum_{k_t=0}^k \sum_{k_x=0}^{k-k_t} \widetilde{\mathbf{U}}_j(k_x, k_t) (x - x_j)^{k_x} (t - t^n)^{k_t}.$$

Evaluating this expression at $t = t^n$ eliminates all terms with $k_t \geq 1$ and gives

$$\mathbf{U}_{\tau,j}(x, t^n) = \sum_{k_x=0}^k \widetilde{\mathbf{U}}_j(k_x, 0) (x - x_j)^{k_x} = \mathbf{U}_{h,j}^n(x).$$

Therefore, the coefficients $\tilde{\mathbf{U}}_j(k_x, 0)$ are precisely the coefficients of the known DG polynomial $\mathbf{U}_{h,j}^n$ expressed in the monomial basis centered at x_j .

At the initial time $t^0 = 0$, these coefficients are obtained from the L^2 -projection of the prescribed initial condition $\mathbf{U}_0$ onto $\mathbb{P}^k(I_j)$. At every subsequent time t^n , $n \geq 1$, they are obtained directly from the previously computed DG polynomial $\mathbf{U}_{h,j}^n$; thus, no new projection of the initial condition is performed.

The predictor $\mathbf{U}_{\tau,j}$ is an auxiliary polynomial used to evaluate the space–time volume integrals, source terms, and time-dependent interface fluxes appearing in the ADER–DG formulation. Since it is constructed locally and independently in each cell, it does not yet include the numerical-flux corrections associated with interactions between neighboring cells. Consequently, although

$$\mathbf{U}_{\tau,j}(x, t^n) = \mathbf{U}_{h,j}^n(x),$$

one generally has

$$\mathbf{U}_{\tau,j}(x, t^{n+1}) \neq \mathbf{U}_{h,j}^{n+1}(x).$$

The updated DG solution $\mathbf{U}_h^{n+1}$ is obtained only after applying the globally coupled DG correction step, in which the local predictors are used to compute the volume contributions and numerical fluxes. The ADER–DG evolution can therefore be summarized as

$$\mathbf{U}_h^n \longrightarrow \mathbf{U}_\tau \longrightarrow \mathbf{U}_h^{n+1},$$

where the first arrow represents the element-local space–time prediction and the second represents the globally coupled DG update.